Q.24 What is explosive welding? Explain the process.

Q.25 What is water hammer process? Discuss its applications.

Q.26 Explain the electro-magnetic forming process.

Q.27 Draw the neat sketch of Laser Jet machining with main parts.

Q.28 Draw abrasive flow machining process.

Q.29 Discuss chemical machining and its drawbacks.

Q.30 What type of jobs are suitable for Electrochemical machining and why?

Q.31 Describe explosive compaction process.

Q.32 Give the tool material suitable of USM, EDM processes giving the reason.

Q.33 Explain diffusion process.

Q.34 Write at least five disadvantages of EBM.

Q.35 What is Chemical Grinding? Explain.

## SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the principle, working and equipment needed for WEDM process with its line diagram.

Q.37 Give the brief description of diffusion and photolithography process for manufacturing electronic devices.

Q.38 Explain with neat sketch E.B.M. Explain its working principle, with mechanism of material removal. Also mention three applications of E.B.M.

(20)





Roll No. ..................

**3rd Sem / Mechatronics**

**Subject:- Non conventional Manufacturing Processing/ Manufacturing Process - I**

Time : 3Hrs. M.M. : 100

## SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 ECM process is based on which of the following laws?

a) Coulomb's law

b) Faraday's law

c) Law of definite proportions

d) Law of chemical combination

Q.2 USM removes material using the______ tool.

a) Perpendicularly rotating

b) Perpendicularly oscillating

c) Axially oscillating

d) Inclined oscillating

Q.3 What is the velocity of water jet stream in water jet machining?

a) 100 m/sec b) 300 m/sec

c) 700 m/sec d) 900 m/sec

Q.4 Material removal take place in AJM due to

a) Electrochemical action

b) Mechanical impact

c) Fatigue failure of the material

d) Sparking on impact

Q.5 In abrasive jet machining, work piece material of removed by which of the following means?

a) vaporization  b) Electro plating
c) Mechanical abrasion  d) Corrosion

Q.6 Mechanism of material removal in Electron Beam Machining is due to

a) Mechanical erosion due to impact of high of energy electrons
b) Chemical etching by the high energy electron
c) Sputtering due to high energy electrons
d) Melting and vaporisation due to thermal effect of impingment of high energy electrons

Q.7 Which is not characteristics of cut and peel maskants in chemical machining?

a) It gives high chemical resistance
b) Depth of etchant is restricted to 13 mm
c) It is not suitable for obtaining critical dimensional tolerances.
d) Re-scribing of maskant is not possible.

Q.8 Which of the following gases is not used in Abrasive jet machining?

a) Air  b) Nitrogen
c) Carbon di-oxide  d) Argon

Q.9 In Electrical Discharge Machining (EDM), the spark gap is kept between____ mm to ____mm.

a) 0.5 to 5  b) 0.05, 0.5
c) 0.005, 0.05  d) 0.0005, 0.005



Q.10 Sources used in thermal machining are_______.

a) Ions  b) Plasma
c) Electrons  d) All of the above

## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define unconventional machining process.

Q.12 In E.C.M tool does not touch the workpiece. (T/F)

Q.13 Give the name of chemical mostly used in chemical machining.

Q.14 Define cladding.

Q.15 High-temperature ionized gas is also known as _____.

Q.16 _____ is a process which involved depositing a thin metallic film on the surface of non-metallic objects.

Q.17 What is photo lithography?

Q.18 Under water welding also known as _____ Welding which involves welding at an elevated pressure.

Q.19 Expand LASER.

Q.20 What is the high energy in forming process?

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Discuss the future of non-conventional machining processes.

Q.22 Explain EDM process with its diagram.

Q.23 Explain the working principle of ECM.